Weak points in the ECDLP hardness corpus: triage, 2026-11-03

Question. Where is the claim graph that supports ECDLP hardness weakest, and which weak points can be broken cheaply enough to attack first?

Scope. The assembled corpus (version 2026-11-01: 541 claims, 909 edges, 233 sources), read through `weakness.mjs`, plus the repository sources the corpus cites. Nothing here changes a claim's status or edges; every proposed change is listed as an action. The generated companion is `WEAKNESSES.md` (all rows) with `weakness.json` and `weakness-queue.svg`.

Method. `attack-graph.mjs` already ranks unestablished claims by impact (how much rests on them) and propagates taint from refuted claims. `weakness.mjs` adds what that pass does not see:

check	what it finds	count
status inflation	proved / algorithm / measured claims whose support closure reaches an open, heuristic or assumed claim	17
unwired dependence	a claim's own assumption text invokes a heuristic node with no edge to it	11
live obstruction	an established claim `obstructs` / `qualifies` / `does_not_cover` a claim that still supports others	49 (17 `obstructs`)
hidden shared assumption	the same assumption text on 3+ nodes, never a node itself	17 clusters
source contagion	a primary source with a refuted claim and other live claims	19 sources
model-scope leap	generic-model or measured claim supporting a broader one without `model_bridge`	9

It then ranks the 219 attackable claims by impact × breakability, where breakability is the expected cost of a decisive result: a heuristic with a stated formula in an algebraic model can be checked exactly on small fields; an open literature conjecture cannot. Established `obstructs` edges multiply the score. Reading a primary text or auditing a kernel-checked statement goes in a separate hygiene queue (91 claims): cheap, but it rarely overturns anything. **The scores are a triage heuristic, not evidence.**

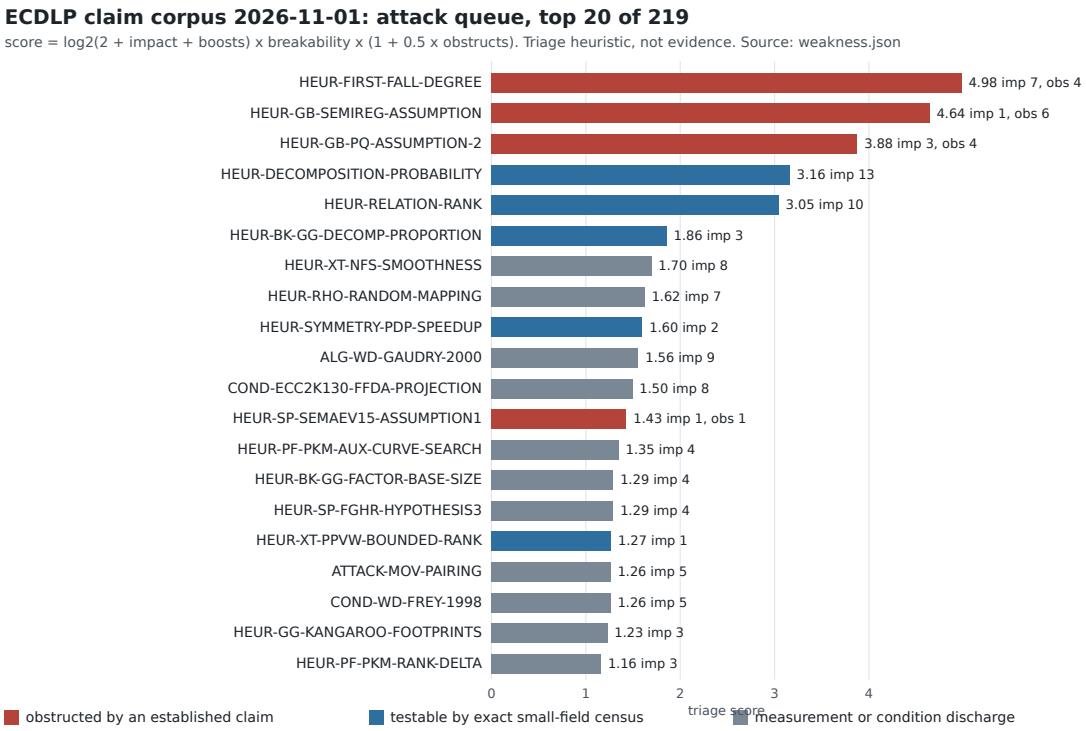


Figure 1: Attack queue, top 20 (weakness-queue.svg)

Evidence labels below: **exact** = a count or graph fact over the corpus files; **read** = taken from a cited repository source, quoted path given; **derived** = an argument written here, not yet reviewed or kernel-checked; **scratch** = a toy check run in this session's scratch space, not committed and not evidence for the corpus.

W1. The first-fall-degree pillar is obstructed and still load-bearing

Exact. `HEUR-FIRST-FALL-DEGREE` has four incoming `obstructs` edges from established claims (CG21 incomparability, Kosters–Yeo's first fall degree 2 on descended S_3 at every n, and two Caminata–Gorla summation measurements), and it still conditions seven claims: `COND-ECC2K130-FFDA-PROJECTION` (the 2^86 projection), `COND-NAGAO-2015-COMPLEXITY`, `COND-PKM-PRIME-ALGEBRAIC`, `COND-BINARY-WEIL-DESCENT`, `COND-GB-HFE-FIRST-FALL-ESTIMATES`, `COND-SP-HPST-SYMMETRIZED-PRIME-DEGREE`, `COND-SP-KARABINA-PDP`. Its siblings are in the same state: `HEUR-GB-SEMIREG-ASSUMPTION` has six `obstructs` edges, `HEUR-GB-PQ-ASSUMPTION-2` four. These three take the top of the attack queue. `HEUR-GB-PQ-ASSUMPTION-2` also states the first-fall assumption in its own text with no edge to `HEUR-FIRST-FALL-DEGREE`.

Judgment. As a general principle the first-fall-degree assumption is already dead in the corpus's own terms. What survives is a family-specific version for each estimate that uses it. The graph keeps it at `heuristic`, so the seven conditional estimates read as one discharged condition away from usable, which overstates them.

Attack. Petit–Quisquater Assumption 2 is asymptotic (`D_reg = D_ff + o(1)`), so no finite table refutes it; the repository's measurements (D_ff = 2, degree 4→5 at n = 36→37, 5 at n = 45) only obstruct it. A **lower-bound proof** settles it: show that the solving degree of the descended S_3 system at m = 2 is ≥ 3 for all $n \geq n_0$. Degrees are integers, so `+ o(1)` means eventual equality, and a uniform lower bound of 3 with D_ff = 2 refutes the assumption for that family. Candidate route: exhibit, for every n, a degree-2 Macaulay span that provably misses a polynomial the reduced Gröbner basis needs. A census of the degree-2 Macaulay rank against n over F_2 would show whether the gap is uniform before the proof is attempted.

Effect on the baseline. None directly. This retracts projected IC costs that were never the measured baseline. It matters because it removes a whole family of "conditional subexponential" rows from the menu of routes worth funding.

W2. The relation-rank heuristic is load-bearing only by a modelling choice

Exact. `HEUR-RELATION-RANK` conditions `ALG-SPARSE-LINEAR-ALGEBRA`, and through it the corpus marks `THM-WD-GTTD-THM1`, `THM-WD-DIEM-SMALL-GENUS` and `ATTACK-GHS-WEIL-DESCENT` (all `proved_external`) as resting on a heuristic: 4 of the 17 status-inflation rows. Five more claims invoke rank in their assumption text with no edge (W5).

Read. `HEUR-BK-GG-RIGHT-KERNEL-RANK` (Galbraith–Gaudry 2016, §6.2) already records that left-kernel (target-dependent) relations "are much easier to analyze rigorously", while right-kernel relations need rank #F − 1. No node carries the left-kernel argument and no edge discharges the rank heuristic with it.

Derived (needs review and a Lean statement). Let r be prime and Q = xP. Draw (a_i, b_i) uniformly, set $R_i = a_iP + b_iQ$, and keep the R_i that decompose over the factor base F, giving rows $M_i \in (Z/r)^{\wedge}F$. Take B = |F| + 1 rows. Some nonzero λ satisfies λM = 0, whatever the rank of M, and then $\Sigma\lambda_i a_i + x \cdot \Sigma\lambda_i b_i \equiv 0 \pmod r$. Fix the points R_i: the decomposition, and so M and λ, depend only on the R_i. Given R_i, b_i is still uniform on Z/r (the r pairs with $a_i + b_i x = \log R_i$ are equally likely). So $\Pr[\Sigma\lambda_i b_i \equiv 0] = 1/r$ and x is recovered with probability $1 - 1/r$. Signed/automorphism-folded columns change nothing. **No rank condition is**

used. That is why GTTD and Diem can be rigorous, and the four inflation rows are edge errors, not fragile theorems.

Scratch. Toy check (p = 10007, prime-order curves, factor base x < 60, two-term decompositions, |F| + 1 relations, 12 seeds): every matrix was rank-deficient (2–7 short of |F|), so the right-kernel formulation fails on every one, and the left-kernel recovery returned the correct log in 12/12. Illustration only.

Where the heuristic is truly load-bearing. Only where relations are reused across targets (right-kernel amortisation): multi-target batches such as the `ic.residual_batch` vs `rho.multitarget_pooled` parity work, and PKM's $\Delta = 10$ surplus (`HEUR-PF-PKM-RANK-DELTA`). There the measurable question is the rank deficit as a function of |F| and the column-folding convention, which a native census can answer exactly.

Actions. (a) Add a `THM-LEFT-KERNEL-RELATIONS` node (derived above; formalise the finite core in Lean: uniformity of b_i given R_i and the 1/r bound). (b) Re-wire the rigorous Weil-descent chain to it, which removes 4 inflation rows. (c) Narrow `HEUR-RELATION-RANK` to the multi-target / right-kernel regime and add the missing edges from W5. (d) Measure rank deficit on the multi-target factor bases the repository already uses.

W3. The decomposition-yield heuristic conflates two regimes

Exact. `HEUR-DECOMPOSITION-PROBABILITY` has the largest impact on the frontier (13), and four more claims invoke it in text without an edge (W5). Its statement already says the mean B^m/r "follows by counting": only the distribution shape (Poisson, variance, independence, rationality, solver visibility) is heuristic.

Judgment. Two different claims share this node. Over F_{q^n} with n fixed, Diem's rigorous results (`THM-WD-DIEM-SMALL-GENUS`; its scope records the rigorous q^(2−2/n) ECDLP bound cited by KMPS) bypass the heuristic. Over prime fields it is genuinely heuristic, and that is where this repository's programme lives. Under the left-kernel formulation (W2), the only quantity an index-calculus cost needs is Pr[a uniform point decomposes] = |{R : R decomposes}| / r, which is a **set size, not a heuristic**, and exact on small fields.

Attack. Split the node into an extension-field version (cite Diem; status follows the read) and a prime-field version. For the prime-field one, run an exact census over every target on a scaling family of small p. Record the full histogram of representation counts, not just the mean, and report the observed decomposable fraction against $1 - \exp(-\mu)$ with its trend in p. A systematic deviation is a

correction to every prime-field yield estimate; agreement is a measured bound. Either outcome is decisive at the measured scale.

W4. The Lemma 4 refutation rests on a reading choice, and two internal papers disagree

Exact. Eleven nodes in the Nagao Lemma 4 family carry the free-text assumption "Equality reading of Definitions 5 and 6", the largest hidden-assumption cluster. `CLAIM-NAGAO-LEMMA4` is `refuted`, and its scope says the refutation is reading-specific.

Read. `papers/ecdlp-boundaries-20261018/sections/fall-degree.tex` tabulates all four (reading × placement) cells. The equality reading is refuted both outside and inside; literal ≥ outside is refuted (d_F = 3, d'_F = 2); literal ≥ inside gives d_F = d'_F = 2. It also states that the wording "the literal reading always voids the counterexample is too broad". But `papers/ic-negative-results/ic_negative_results.tex` (≈ line 470), the repository source for the refutation (`S-IC-NEGATIVE`), still says "under that literal reading the lemma holds only vacuously". The boundaries section also notes that a final publication "still owes a fresh PDF-level transcription of both definitions".

Actions. (a) Correct the ic-negative wording to the boundaries table. (b) Record the reading as a parameter of `CLAIM-NAGAO-LEMMA4` (refuted in 3 of 4 cells) instead of a footnote. (c) Run the existing exact census on the one open cell (literal ≥, field equations inside). (d) Do the PDF transcription, because the refutation is only as good as the reading.

W5. Eleven unwired dependences understate the top frontier nodes

Exact. These claims name a heuristic in their own assumptions but have no support path to its node, so refuting the heuristic would not reach them:

- decomposition yield → `ALG-GAUDRY-FIXED-EXTENSION`, `COND-BINARY-WEIL-DESCENT`, `COND-GB-PQ-SUBEXP-BINARY-ECDLP`, `COND-SP-KARABINA-PDP`
- relation rank → `COND-PF-PKM-COST-THRESHOLD`, `ALG-GAUDRY-FIXED-EXTENSION`, `COND-BINARY-WEIL-DESCENT`, `COND-GB-PQ-SUBEXP-BINARY-ECDLP`, `COND-PKM-PRIME-ALGEBRAIC`, `HEUR-XT-NFS-SMOOTHNESS` (NFS rank is the same phenomenon in a finite-field setting; wire it or record why not)
- first fall degree → `HEUR-GB-PQ-ASSUMPTION-2`

Action. Add the `conditions` edges, then re-run `attack-graph.mjs`. `weakness.mjs` now reports any new occurrence.

W6. Smaller structural findings

- **Conditional obstruction labelled proved (exact).** `OBSTRUCTION-XEDNI-HEIGHT` is `proved_external`, but it rests on Lang's height conjecture (`CONJ-XT-LANG-HEIGHT`, open) and six more hypotheses its own scope lists. The xedni line is closed only conditionally. Relabel it `conditional`. As an attack route it is still poor (it needs a uniform failure of Lang's conjecture on the selected lifts).
- **In-model theorems resting on an assumption (exact).** Five rho theorems (KMPT, KMT, MV mixing and nondegeneracy, three-partition) are `proved_external` given `ASM-GG-RANDOM-PARTITION`. That is correct inside the model, but implemented r-adding walks are not random partitions. This affects the **rho reference**, not IC: the repository already matches rho empirically, which is the right control.
- **Source contagion (exact).** Among primary sources, Semaev 2015 has 2 of its claims refuted while `HEUR-SP-SEMAEV15-ASSUMPTION1` ($d_{F4} \le 4$) is still live and already obstructed once (Kosters–Yeo: degree 5 at $n = 45$). `RT-SP-SEMAEV-A1-N45` is the decisive test and is cheap. Kosters–Yeo, Euler–Petit/HKPYY and BGJT each have one refuted claim and live siblings (listed in `WEAKNESSES.md` §4).
- **Model-scope leaps (exact).** All nine are generic-model rho heuristics feeding standards or the concrete rho algorithm. That is the honest, well-known generic-to-concrete step, and the repository's matched-rho measurements control it.

Recommended order of attack

#	target	kind of result	cost	what changes if it lands
1	W2: left-kernel lemma + Lean core; re-wire; narrow rank heuristic	proof	low	4 inflation rows resolved; rank heuristic confined to the multi-target regime
2	W5: add the 11 missing edges	corpus fix	trivial	true impact of the top frontier nodes becomes visible
3	W4: fix the ic-negative wording; census the open cell	correction + exact census	low	the Lemma 4 refutation becomes reading-complete
4	W3: prime-field yield census, full histogram, scaling in p	exact measurement	low–medium	every prime-field yield estimate is checked or corrected
5	W1: degree-2 Macaulay rank census, then the lower-bound proof	census, then proof	medium	PQ Assumption 2 refuted for $m = 2$; the FFDA rows retire
6	`RT-SP-SEMAEV-A1-N45`	exact F4 run	low	Semaev 2015 Assumption 1 settled at the measured scale

Items 1, 2 and 3 are corpus and proof work, not experiment runs. Items 4–6 are runs and go through the repository's experiment route (protocol, resume checkpoint, native code, validation gate) before any number enters the corpus.

Reproduce

```
node docs/ecdlp-hardness-corpus/weakness.mjs          # writes WEAKNESSES.md,
weakness.json, weakness-queue.svg
node docs/ecdlp-hardness-corpus/weakness.mjs --check  # fails if they are stale
```